MAHATMAGANDHI UNIVERSITRY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION 2025
THIRD SEMESTER
FoodTechnologyandQualityAssurance
MG3DSCFTQ202:UNITOPERATIONSINFOODINDUSTRY

(2024 ADMISSION ONWARDS)

Duration: **2 Hours** **Maximum Marks: 50**

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2**marks

1. Write about cryogenic freezers. [U]/[CO1]
2. Explain the difference between direct and indirect heat exchange with suitable examples. [U]/[CO1]
3. What is reverse osmosis. [E]/[CO4]
4. What are the advantages of freeze drying. [U]/[CO2]
5. Define evaporation and types of evaporators. [A]/[CO3]
6. How sedimentation is used in separation of food components? [C]/[CO5]
7. What are the different types of driers in food industry? [U]/[CO2]

[5X2 =10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

8. Define Food freezing. Explain the principle behind the freezing process. [U]/ [CO1]
9. Explain the mechanism of filtration and how it separates solids from liquids in food processing. [E] /[CO4]

10. Explain shell and tube heat exchanger with a neat and labelled sketch. [A]/[CO3]

11. Explain Newton's law of cooling. [U]/[CO1]

12. Define clarification and sedimentation. Write a detailed note on sedimentation equipment and its applications. [C]/[CO5]

13. Explain how steam infusion and steam injection systems enhance heat transfer in fluid foods. [A] / [CO3]

14. Describe the mechanism of water distillation . [E] / [CO4]

[5X4= 20]

Part C

Answer any **2** questions

Each question carries **10** marks

15. Discuss the principles of food freezing and list the factors affecting the freezingprocess. [U] / [CO1]

16. Describe the construction and working of spray dryer, fluidized bed dryer, andfreeze dryer. [A]/[CO2]

17. Discuss sedimentation and how it is used along with clarification in separationprocesses. [A]/[CO5]

18. Describe the theory of filtration and explain the working of plate and frame filterpress. [U]/[CO4]

[2X10= 20]